2.66 Numerical Methods: Improved Euler's Method

Worked Solutions

1 a)

$$\begin{aligned} x_0 &= 1, y_0 = 2 \\ y_1 &= 2 + 0.5\text{f}(x_0, y_0) \\ &= 2 + 0.5 \times 0 = 2 \\ x_1 &= 1 + 0.5 = 1.5 \\ \therefore y(2) &= 2 \end{aligned}$$

1 b)

$$\begin{aligned} y_2 &= y_0 + 2h\text{f}(x_1, y_1) \\ &= 2 + 2 \times 0.5 \times 0.8047189562\ldots = 2.804718956\ldots \\ x_2 &= 1.5 + 0.5 = 2 \\ \therefore y(2) &= 2.805 \text{ (3 DP)} \end{aligned}$$

2)

$$\begin{aligned} x_0 &= 2, y_0 = 1 \\ y_1 &= 1 + 0.1\text{f}(x_0, y_0) \\ &= 1 + 0.1 \times 2.693147181\ldots = 1.269314718\ldots \\ x_1 &= 2 + 0.1 = 2.1 \\ y_2 &= y_0 + 2h\text{f}(x_1, y_1) \\ &= 1 + 2 \times 0.1 \times 2.9194779\ldots = 1.58389558\ldots \\ x_2 &= 2.1 + 0.1 = 2.2 \\ \therefore y(2.2) &= 1.584 \text{ (3 DP)} \end{aligned}$$

3 a)

$$\begin{aligned} x_0 &= 0, y_0 = 1 \\ x_1 &= 0.75, y_1 = 1.75 \\ y_2 &= y_0 + 2h\text{f}(x_1, y_1) \\ &= 1 + 2 \times 0.75 \times 2.943137453\ldots = 5.41470618\ldots \\ \therefore y(1.5) &= 5.415 \text{ (3 DP)} \end{aligned}$$

3 b)

$$
\begin{aligned}
x_0 &= 1, y_0 = 0.2 \\
x_1 &= 1.25, y_1 = 0.2779 \\
y_2 &= y_0 + 2h\mathrm{f}(x_1, y_1) \\
&= 0.2 + 2 \times 0.25 \times 0.8363594124\ldots = 0.6181797062\ldots \\
x_2 &= 1.25 + 0.25 = 1.5 \\
y_3 &= y_1 + 2h\mathrm{f}(x_2, y_2) \\
&= 0.2779 + 2 \times 0.25 \times 8.71721164\ldots = 4.63650582\ldots \\
\therefore y(1.75) &= 4.637 \text{ (3 DP)}
\end{aligned}
$$

4 a)

$$
\begin{aligned}
x_0 &= 1, y_0 = 3 \\
k_1 &= 2.633916794\ldots \\
k_2 &= 4.2501647216535\ldots \\
x_1 &= 3, \quad y_1 = 3 + \frac{1}{2}(2.633916794\ldots + 4.2501647216535\ldots) \\
&= 6.442040258\ldots \\
\therefore y(3) &= 6.44
\end{aligned}
$$

4 b)

$$
\begin{aligned}
x_0 &= 2, y_0 = 0.5 \\
k_1 &= 0.2459397694\ldots \\
k_2 &= 0.3321224914\ldots \\
x_1 &= 2.2, \quad y_1 = 0.5 + \frac{1}{2}(0.2459397694\ldots + 0.3321224914\ldots) \\
&= 0.7890461304\ldots \\
\therefore y(2.2) &= 0.789
\end{aligned}
$$

4 c)

$$
\begin{aligned}
x_0 &= 0, y_0 = 3 \\
k_1 &= 1.732050808\ldots \\
k_2 &= 4.002893403\ldots \\
x_1 &= 3, \quad y_1 = 3 + \frac{1}{2}(1.732050808\ldots + 4.002893403\ldots) = 5.867472105\ldots \\
\therefore y(0.5) &= 5.87
\end{aligned}
$$

5 a)

$$\begin{aligned}
x_0 &= 4, y_0 = e^{0.1} \\
y_1 &= e^{0.1} + 0.1\mathrm{f}(x_0, y_0) \\
&= e^{0.1} + 0.1 \times 3.315512754\ldots = 1.436722193\ldots \\
x_1 &= 4 + 0.1 = 4.1 \\
y_2 &= y_0 + 2h\mathrm{f}(x_1, y_1) \\
&= e^{0.1} + 2 \times 0.1 \times 4.357794176\ldots = 1.97672953\ldots \\
x_2 &= 4.1 + 0.1 = 4.2 \\
\therefore y(4.2) &= 1.977 \text{ (3 SF)}
\end{aligned}$$

5 b)

$$\int \frac{1}{y}\,dy = \int \sqrt{2x+1}\,dx = \int (2x+1)^{1/2}\,dx$$

$$\ln y = \frac{1}{3}(2x+1)^{3/2} + c$$

Given $(4, e^{0.1})$ then

$$\ln(e^{0.1}) = \frac{1}{3}(2 \times 4 + 1)^{3/2} + c \Rightarrow c = -8.9$$

$$\therefore \ln y = \frac{1}{3}(2x+1)^{3/2} - 8.9$$

5 c)

$$\frac{1.977 - e^{\frac{1}{3}(2\times 4.2+1)^{3/2}-8.9}}{e^{\frac{1}{3}(2\times 4.2+1)^{3/2}-8.9}} = -0.025$$

6 a)

$$\begin{aligned}
x_0 &= 0, y_0 = 1 \\
y_1 &= 1 + 0.05\mathrm{f}(x_0, y_0) \\
&= 1 + 0.05 \times -\frac{1}{2} = 0.975 \\
x_1 &= 0 + 0.05 = 0.05 \\
y_2 &= y_0 + 2h\mathrm{f}(x_1, y_1) \\
&= 1 + 2 \times 0.05 \times -0.4375 = 0.95625 \\
x_2 &= 0.05 + 0.05 = 0.1 \\
\therefore y(0.1) &= 0.95625
\end{aligned}$$

6 b)

$$\frac{dy}{dx} + \frac{1}{2}y = x$$

I.F. is $e^{\int \frac{1}{2}\,dx} = e^{x/2}$

$$e^{x/2}\frac{dy}{dx} + \frac{1}{2}e^{x/2}y = xe^{x/2}$$
$$e^{x/2}y = \int xe^{x/2}\,dx$$

Use Integration by Parts:

$$u = x, \quad \frac{dv}{dx} = e^{x/2}, \quad \frac{du}{dx} = 1, \quad v = 2e^{x/2}$$

$$\therefore \int xe^{x/2}\,dx = 2xe^{x/2} - \int 2e^{x/2}\,dx = 2xe^{x/2} - 4e^{x/2}(+c)$$

So that:

$$e^{x/2}y = 2xe^{x/2} - 4e^{x/2} + c$$
$$y = 2x - 4 + ce^{-x/2}$$

6 c)

Given $(0, 1)$ then

$$1 = 0 - 4 + c \Rightarrow c = 5$$
$$\therefore y = 2x - 4 + 5e^{-x/2}$$

When $x = 0.1, y = 2 \times 0.1 - 4 + 5e^{-0.1/2} = 0.956147\ldots$

$$\therefore y(0.1) = 0.956 \text{ (3 SF)}$$

6 d)

The value from part a) is equal to the answer in part c) to three significant figures, meaning the estimate in part a) is very accurate.

7 a)

$$\begin{aligned}\frac{dC}{dt} &= -0.2C + 1.5e^{-0.1t} \\ t_0 &= 0, C_0 = 0 \\ C_1 &= 0 + 0.5\text{f}(t_0, C_0) \\ &= 0 + 0.5 \times 1.5 = 0.75 \\ \therefore C &= 0.75 \text{ mg/L}\end{aligned}$$

7 b)

A greater number of smaller steps could be used.

7 c)

$$
\begin{aligned}
t_0 &= 0, C_0 = 0 \\
k_1 &= hf(x_r, y_r) = 0.5f(t_0, C_0) = 0.5 \times 1.5 = 0.75 \\
k_2 &= hf(x_r + h, y_r + k_1) = 0.5f(0.5, 0.75) = 0.6384220684\ldots \\
y_{r+1} &= y_r + \frac{1}{2}(k_1 + k_2) \\
C_1 &= 0 + \frac{1}{2}(0.75 + 0.6384220684\ldots) \\
&= 0.6942110342\ldots \\
t_1 &= 0 + 0.5 = 0.5 \\
\therefore C &= 0.69 \text{ mg/L}
\end{aligned}
$$

8 a)

$$(15 - 0.04t^2)\frac{dT}{dt} = 6t - 0.8T \Rightarrow \frac{dT}{dt} = \frac{6t - 0.8T}{15 - 0.04t^2}$$

$$
\begin{aligned}
t_0 &= 0, T_0 = 20 \\
T_1 &= 20 + 0.5f(t_0, T_0) \\
&= 20 + 0.5 \times -\frac{16}{15} = 19.46666667\ldots \\
t_1 &= 0 + 0.5 = 0.5 \\
T_2 &= 19.46666667\ldots + 0.5f(t_1, T_1) \\
&= 19.46666667\ldots + 0.5 \times -0.8387814098\ldots = 19.04727596 \\
t_2 &= 0.5 + 0.5 = 1 \\
\therefore T &= 19.0^\circ\text{C}
\end{aligned}
$$

8 b)

A greater number of smaller steps could be used.

8 c)

$$
\begin{aligned}
t_0 = 0, T_0 &= 20 \\
t_0 + \frac{h}{2} &= 0 + \frac{1}{2} = 0.5 \\
T_0 + \frac{h}{2}f(t_0, T_0) &= 20 + \frac{1}{2} \times -\frac{16}{15} = 19.46666667\ldots \\
\therefore T_1 &= T_0 + hf\left(t_0 + \frac{h}{2}, T_0 + \frac{h}{2}f(t_0, T_0)\right) \\
&= 20 + 1 \times f(0.5, 19.46666667\ldots) \\
&= 20 + 1 \times -0.8382222222\ldots
\end{aligned}
$$

$$= 19.161777777\ldots$$
$$\therefore T = 19.2^{\circ}\text{C}$$

9)

$$x_0 = 0, y_0 = 1$$
$$k_1 = \text{f}(x_0, y_0) = \text{f}(0, 1) = 0 - 1 = -1$$
$$k_2 = \text{f}(x_0 + 0.25, y_0 + 0.25 \times -1) = \text{f}(0.25, 0.75) = 0.25 - 0.75 = -0.5$$
$$k_3 = \text{f}(x_0 + 0.25, y_0 + 0.25 \times -0.5) = \text{f}(0.25, 0.875) = -0.625$$
$$k_4 = \text{f}(x_0 + 0.5, y_0 + 0.5 \times -0.625) = \text{f}(0.5, 0.6875) = -0.1875$$
$$y_1 = 1 + \frac{1}{12}(-1 + 2(-0.5) + 2(-0.625) - 0.1875) = 0.713541666\ldots$$
$$\therefore y = 0.714 \text{ (3 DP)}$$

10)

$$h = 0.1$$
$$x_0 = 1, y_0 = a$$
$$y_1 = a + 0.1\text{f}(x_0, y_0) = a + 0.1 \times (5 - a) = 0.9a + 0.5$$
$$x_1 = 1 + 0.1 = 1.1$$
$$y_2 = y_0 + 2h\text{f}(x_1, y_1) = a + 0.2 \times (5.5 - (0.9a + 0.5)) = 0.82a + 1$$
$$\Rightarrow 0.82a + 1 = 3.05$$
$$\therefore a = 2.5$$

11 a)

$$x_0 = 1, y_0 = 0.5$$
$$y_1 = 0.5 + h\text{f}(x_0, y_0) = 0.5 + h \times (1 - 0.5) = 0.5h + 0.5$$

Since $y_1 = 0.6$ then $0.5h + 0.5 = 0.6$

$$\therefore h = 0.2$$

11 b)

$$x_1 = 1 + 0.2 = 1.2$$
$$y_2 = y_0 + 2h\text{f}(x_1, y_1) = 0.5 + 2 \times 0.2(1.2^2 - 0.6) = 0.836$$
$$x_2 = 1.2 + 0.2 = 1.4$$
$$\therefore (x_1, y_1) = (1.4, 0.836)$$

12)

$$
\begin{aligned}
&x_0 = 0, y_0 = 1 \\
&y_1 = 1 + h\mathrm{f}(x_0, y_0) = 1 + h \times (0 + 1) = 1 + h \\
&x_1 = 0 + h = h \\
&y_2 = 1 + 2h\mathrm{f}(x_1, y_1) = 1 + 2h \times (1 + h + h) = 4h^2 + 2h + 1
\end{aligned}
$$

Since $y_2 = 1.96$ then $4h^2 + 2h + 1 = 1.96 \Rightarrow 4h^2 + 2h - 0.96 = 0$

$$\Rightarrow h = -0.8 \text{ or } h = 0.3$$

$$\therefore h = 0.3$$

13)

$$
\begin{aligned}
h &= 0.5 \\
x_0 &= 2, y_0 = a \\
k_1 &= 0.5(2a - 6) = a - 3 \\
k_2 &= 0.5(2(a + a - 3) - 3(2 + 0.5)) = 2a - 6.75 \\
y_1 &= a + \frac{1}{2}(a - 3 + 2a - 6.75) = 2.5a - 4.875 \\
\therefore y(2.5) &= 7.625 \Rightarrow 2.5a - 4.875 = 7.625 \\
\therefore a &= 5
\end{aligned}
$$

14)

$$
\begin{aligned}
x_0 &= 0, y_0 = 1 \\
k_1 &= h(4 + 0) = 4h \\
k_2 &= h(4(1 + 4h) + 2h) = 18h^2 + 4h \\
y_1 &= 1 + \frac{1}{2}(4h + 18h^2 + 4h) \\
\therefore 1 + \frac{1}{2}(4h + 18h^2 + 4h) &= 2.16 \Rightarrow 9h^2 + 4h - 1.16 = 0 \\
&\Rightarrow h = -0.6444444\ldots \text{ or } h = 0.2 \\
&\therefore h = 0.2
\end{aligned}
$$